

Centavo — Database Design

Schema

users

Column	Definition
id	INT PRIMARY KEY AUTO_INCREMENT
email	VARCHAR(255) UNIQUE NOT NULL
password_hash	VARCHAR(255) NOT NULL
name	VARCHAR(100) NOT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

accounts

Column	Definition
id	INT PRIMARY KEY AUTO_INCREMENT
user_id	INT NOT NULL, FOREIGN KEY -> users(id) ON DELETE CASCADE
name	VARCHAR(100) NOT NULL
type	ENUM('checking','savings','credit_card','cash') NOT NULL
balance	DECIMAL(12,2) NOT NULL DEFAULT 0
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Money is stored as DECIMAL(12,2), never FLOAT. Floating point introduces rounding error in financial math that looks fine in testing and quietly corrupts balances later. MySQL forces this decision explicitly.

categories

Column	Definition
id	INT PRIMARY KEY AUTO_INCREMENT
user_id	INT NULL, FOREIGN KEY -> users(id) ON DELETE CASCADE
name	VARCHAR(100) NOT NULL
is_default	BOOLEAN NOT NULL DEFAULT FALSE
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

user_id NULL indicates a system default category, shared across all users.

transactions

Column	Definition
id	INT PRIMARY KEY AUTO_INCREMENT
user_id	INT NOT NULL, FOREIGN KEY -> users(id) ON DELETE CASCADE

Column	Definition
type	ENUM('expense','income','transfer') NOT NULL
amount	DECIMAL(12,2) NOT NULL
date	DATE NOT NULL
note	VARCHAR(255)
category_id	INT NULL, FOREIGN KEY -> categories(id)
account_id	INT NOT NULL, FOREIGN KEY -> accounts(id)
to_account_id	INT NULL, FOREIGN KEY -> accounts(id)
savings_goal_id	INT NULL, FOREIGN KEY -> savings_goals(id)
source	ENUM('manual','ai_text') NOT NULL DEFAULT 'manual'
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Indexes:

- INDEX (user_id, category_id, date) — covers category-trend and anomaly-detection queries.
- INDEX (user_id, account_id) — for account-statement views.
- INDEX (user_id, date) — for general date-range dashboard queries.

budgets

Column	Definition
id	INT PRIMARY KEY AUTO_INCREMENT
user_id	INT NOT NULL, FOREIGN KEY -> users(id) ON DELETE CASCADE
category_id	INT NOT NULL, FOREIGN KEY -> categories(id)
monthly_limit	DECIMAL(12,2) NOT NULL
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP

UNIQUE (user_id, category_id) — enforces one budget per category per user at the database level, rather than relying on application code to never violate it. Recurring, no history: editing a limit changes it retroactively for all past and future analytics, by design.

savings_goals

Column	Definition
id	INT PRIMARY KEY AUTO_INCREMENT
user_id	INT NOT NULL, FOREIGN KEY -> users(id) ON DELETE CASCADE
name	VARCHAR(100) NOT NULL
target_amount	DECIMAL(12,2) NOT NULL
target_date	DATE NOT NULL
current_amount	DECIMAL(12,2) NOT NULL DEFAULT 0
created_at	TIMESTAMP DEFAULT CURRENT_TIMESTAMP